Semantic Trajectories of Evolving Software Objects

Giovanni Pezzin
Physis Research Program

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Abstract

Revisions of a single source file form a low-dimensional directed semantic trajectory in embedding space. We analyze six files from a Rust codebase and find that (1) within-object distances are 4–14× smaller than between-object distances, (2) five of six files show temporal direction (Spearman ρ up to +0.976, outside a 2000-permutation shuffled-time null), and (3) the trajectories occupy 2–3 dimensions of the 768-dimensional embedding space (57–82% of variance in the first two components). A sixth file (`quality.rs`) fails the direction test because it wanders and returns — an honest failure mode that reveals the limits of trajectory analysis. Document-level identity survives; paragraph-level identity dissolves.

1 Introduction

Software objects evolve. A source file accumulates revisions; each revision is a state of the object at a point in time. The question is whether these states form a *trajectory* — a continuous, directed path through representation space — or a set of independent points.

This question has practical implications. If revisions form a trajectory, then (a) semantic search should use trajectory queries, not point queries; (b) change prediction should extrapolate the trajectory, not predict from a static embedding; (c) object identity should be defined by trajectory proximity, not point proximity.

We test this on six files from a Rust codebase, using BGE-base embeddings (768 dimensions, `onnx-explicit` backend). We measure three properties: identity (within-object distance vs. between-object distance), direction (Spearman correlation of drift with time, against a shuffled-time null), and structure (spectral concentration of the trajectory).

2 Method

For each file f, we:

1. Extract all revisions $r_1, \ldots, r_n$ from git history.
2. Embed each revision (chunked at 1500 characters, mean-pooled into a single 768-d vector).
3. Compute drift $d_i = \|v_i - v_1\|$ (distance from the first revision).
4. Compute Spearman correlation $\rho(i, d_i)$ against a 2000-permutation shuffled-time null.
5. Compute spectral concentration: the fraction of variance in the first two eigenvalues of the centered Gram matrix.

3 Data

Six files from a Rust codebase (`physis-pro`), spanning 8–27 revisions each:

File	Revisions	Drift range	Spearman ρ	2D var
`hypothesis.rs`	12	0.026	+0.930	0.766
`main.rs`	27	0.116	+0.626	0.621
`studio.rs`	17	0.037	+0.939	0.761
`act.rs`	8	0.053	+0.976	0.817
`ontology.rs`	8	0.043	+0.929	0.740
`quality.rs`	9	0.025	+0.250	0.569

Table 1: Trajectory analysis of six source files. Drift range is the difference between maximum and minimum distance from the first revision. 2D var is the fraction of variance explained by the first two eigenvalues of the Gram matrix.

4 Results

4.1 Identity

Within-object distances (mean pairwise cosine distance between revisions of the same file) are 4–14× smaller than between-object distances (mean cosine distance between revisions of different files). A file's revisions form a tight region of the embedding field; a different file is elsewhere.

4.2 Direction

Five of six files show temporal direction: Spearman ρ ranges from +0.626 to +0.976, all outside the 2000-permutation shuffled-time null (which spans approximately [-0.4, +0.7] for most files). The sixth file, `quality.rs`, fails: $\rho = +0.250$, inside the null. Its drift profile shows why: it wanders and comes back, so distance-from-origin stops being a clock.

4.3 Structure

The first two eigenvalues of the centered Gram matrix carry 0.569–0.817 of the variance. These trajectories are 2–3-dimensional objects living in a 768-dimensional space: a spectrum with a knee, not noise.

4.4 Granularity

At document level, identity survives: the document's pooled embedding is closer to its own sections than to another document's sections (nesting margin 0.02–0.03). At paragraph level, identity dissolves: a paragraph of one document is nearly as far from another paragraph of the same document (0.379) as from a paragraph of a different document (0.405).

Positional direction — whether a text moves away from its opening as it proceeds — is a property of genre, not of text. A developmental document (the world-map machine plan) shows strong positional direction at section level ($\rho = +0.665$); a reference document (WHAT_PHYSIS_IS) does not ($\rho = +0.083$).

5 Discussion

Software objects are trajectories. The evidence supports the claim that revisions of a single file form a low-dimensional directed trajectory, not a set of independent points. The trajectory is 2–3 dimensional, temporally directed (in 5/6 cases), and bounded (within-object distance 4–14× smaller than between-object).

The failure mode is informative. `quality.rs` fails the direction test because it wanders and returns. This is the honest failure mode of trajectory analysis: when an object's evolution is not monotonic, distance-from-origin is not a clock. The failure is visible in the same table as the successes, not averaged away.

Identity dissolves below document granularity. A document's paragraphs are not a tighter cluster than paragraphs from different documents. Whatever "the same object" means at revision scale, it is not recoverable from a paragraph. This is consistent with the E80 finding that pooling a project into one vector destroys signal.

Limits. One repository, one embedder (BGE-base-en-v1.5), six files, one metadata axis (commit time). The trajectory instrument is available as `physis-core/examples/reference_spectrum.rs`; the harness is `benchmarks/conatus/e78_reference_spectrum.py`. The expansion to thousands of files across public repos (the P10 experiment in the portfolio) is the highest-priority external replication.